



Exploring, Explaining, and Extending ML & DL Models

Prompt Tuning and Instruction Fine Tuning

Hybrid Fine Tuning Framework for Efficient Language Model Adaptation Combining Prompt Tuning and Instruction Fine Tuning

IT3071 - Machine Learning and Optimization Methods

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Abstract

Recently Large Language Models (LLMs) have significantly advanced on Natural Language Processing. It is visible in the example LLMs like GPT, PaLM, LLaMA and T5. These models have achieved high performance levels on various tasks even the tasks that were previously deemed unattainable. However, traditional fine tuning methods requires large computational power and expenses, and may not generalize well to various tasks.

This report proposes a novel ‘Hybrid Fine Tuning Framework for Efficient Language Model Adaptation Combining Prompt Tuning and Instruction Fine Tuning’ to improve the efficiency, task generalization and scalability of traditional methods. This new proposed model learns prompts based on task formulation instructions, while tuning parameters based on meta-instruction, thereby increasing task adaptability at lower computational overhead. This framework allows dynamically optimized prompts based on instruction embeddings, enabling generalization of LLM to unseen tasks while maintaining robustness against prompt variance. The hybridized method proposes scalability for domain adaptation of LLMs to specialize in knowledge application fields when large, labeled datasets are involved. The method includes iterative prompt generation, embedding alignment with instruction-based objectives, and selective parameter tuning of layers sensitive to instruction guidance. Initial benchmarking on NLP tasks demonstrates valuable performance gains compared to standalone prompt-tuning and instruction fine-tuning approaches. This framework offers a practically scalable solution for raising the bar in LLM performance in research and industry environments. The combination of prompt and instruction fine-tuning is considered one of the most optimistic pathways towards improved adaptability and efficiency of LLMs.

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List of Abbreviations

LLM	Large Language Model
NLP	Natural Language Processing
PT	Prompt Tuning
IFT	Instruction Fine Tuning
T5	Text-to-Text Transfer Transformer
FLAN	Fine-tuned LAnguage Net
LLaMa	Large Language Model Meta AI
GPT	Generative Pre-trained Transformer
T0	T-Zero
PaLM	Pathways Language Model

1. Introduction

In the past few years, Large Language Models like GPT, PaLM, T5, and LLaMA revolutionized the field of Natural Language Processing (NLP) due to the unprecedented success of these models in a wide range of tasks, including reasoning, question answering, summarization, and code writing.

LLMs like GPT, T5 and LLaMa have vast knowledge and they first pre-trained largescale on massive corpora using general objectives such as causal language modelling, masked language modelling and next token prediction. In this stage model is trained by using a large scale of tokens, sourced from sources like books, journals, articles, websites and code repositories, exhausting months of computation and extremely excessive cost. This allows the models to gain general natural language processing skills and full contextual knowledge.

However, to act on the tasks assigned these LLMs must be trained to behave in specific ways depending on the task, if not it may provide generic responses instead of following the exact instructions provided deviating from the expected outcome. Therefore, to reshape these models to follow specific tasks or instructions fine tuning is required. Traditional full fine-tuning focuses on to train on a task specific dataset, retraining the whole model by updating all the parameters for each task. This is expensive, require excessive computational power and memory requirements, resource intensive and difficult to scale. A major disadvantage of this method is is it is prone to causing catastrophic forgetting which means the model forgetting previously learned knowledge.

Prompt Tuning (PT) is a parameter efficient fine-tuning method. In this technique instead of retraining entire model and updating all the model parameters, it freezes the pretrained model and focuses on designing task-specific small set of continuous embeddings called "prompts". Then the model learns what a good response is like for specific tasks and guide the model to produce the desired output. This is a fast and efficient tuning method with minimal storage requirements. Research such as: Prefix-Tuning [10], Prompt Tuning [9], and P-Tuning v2 [4] discusses the improved performance of PT against the traditional full fine tuning. Customer service bots, content generation, medical Q&A system and code completion tools are examples of real-world applications for prompt tuning. In contrast to the advantages, prompt tuning has limitations such as lack of interpretability, lack of instruction understanding, prompt length sensitivity, task specificity, and performance gap. Furthermore, PT often lacks semantic robustness: its learned soft prompts may not generalize well to diverse tasks or invincible instructions, as they are not trained to understand or reason with instructions.

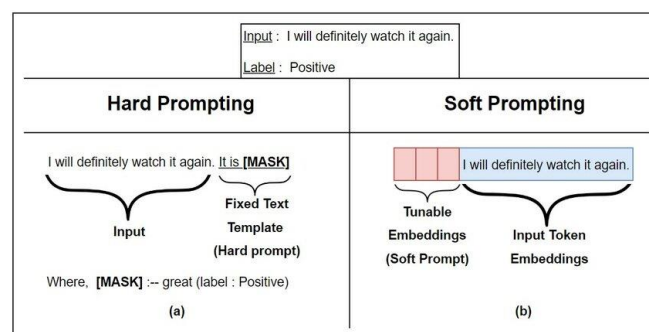


Figure 1-1 Prompt Tuning

Instruction Fine Tuning (IFT) is a specialized form of fine-tuning designed to align a model's behavior with human instructions. It uses a dataset with instruction-output pairs of data also known as “instruction-based” data. This tuning technique learns to map the instructions with the appropriate output. Instruction fine-tuning makes the model follow instructions rather than just following patterns. Improved generalization, enhanced interpretability, reduced risk of producing incorrect or misleading answers, and improved usability are the objectives achieved through implementing IFT. Research such as T0 [11], FLAN [12], InstructGPT [13] are examples for the implementation of Instruction fine-tuning on LLMs. ChatGPT, code assistants and writing assistants are examples of real-world applications of instruction fine tuning. However, the uses of instruction fine tuning are limited due to the requiring large-scale data, GPU resources, and time, making it difficult to continuously adapt or personalize models. It also has limited flexibility, and storage overhead.



Figure 1-2 Instruction Fine Tuning

Considering that Instruction fine-tuning and Prompt tuning complements each other's strengths and weaknesses, combining them provides a new pathway to improve the tuning of the LLMs. The new proposed method will use Instruction fine-tuning to provide the ability of general instruction-following while the use of Prompt tuning will increase the model's efficiency and adaptability. The integration of the new proposal of Hybrid Fine Tuning provides improved instruction understanding, higher efficiency, better performance, flexibility, and cost effectiveness.

1.1 Problem Statement

Thanks to the Instruction fine-tuning LLMs have achieved high performance on tasks they have never experienced before without requiring to train on new examples (zero-shot and few-shot performance). In which they were trained on different types of tasks created as natural language instructions. However, IFT expensive, requires higher computational power, data intensive and not parameter efficient.

In contrast, Prompt tuning is parameter efficient method as it utilizes small set of soft prompts while keeping the model frozen, which in return reduces the cost, storage requirements and computational power. In prompt tuning because the pre-training focuses on specific tasks it often struggles to following instructions across different tasks compared to other models. Therefore, there is requirement of a framework where it is both parameter efficient and improves commitment to instructions.

1.2 Motivation

Regardless of the significant achievements in prompt tuning and instruction fine-tuning, the issue of achieving both efficiency and instruction understanding within the same model is yet to be addressed. While existing Instruction fine tuning methods performs well on tasks following the instructions, they are expensive and require extensive computational power. In contrast, Prompt tuning methods offer efficiency but suffer from semantic understanding and adaptability across various instructions.

This difference motivates the study on a Hybrid Fine-Tuning Framework (HFT) that unites these two primary models: Prompt tuning and Instruction fine-tuning. The fundamental concept is to gain the global instruction understanding through IFT and convert that information into a prompt-based framework for successful efficient task adaptation. This combination would allow models to quickly pick up new actions or follow newly observed instructions without all-parameters retrains efficiently.

This work is motivated on three categories:

- **Practical Motivation:** Adapt tuning to LLMs within affordable cost which reaches smaller organizations and researchers with limited hardware resources.
- **Scientific Motivation:** Investigate if instruction-conditioned representations can guide or extend soft prompt optimization, enhancing task generalization.
- **Sustainability Motivation:** Provide an environmentally friendly AI practice by minimizing the computational and energy footprint associated with full fine-tuning.

If implemented correctly, this proposed hybrid framework would significantly enhance the flexibility, scalability, and sustainability of modern LLMs. It would enable modular updates on single tasks or domains, decrease model storage redundancy, and maintain instruction-following quality key goals in future AI deployment.

2. Literature Review

2.1 Introduction

This literature review explores the recent adaptation in large language models (LLMs) that perform different approaches to respond to human instructions better using models of Prompt Tuning and Instruction Fine Tuning. Most of the existing works are focused on separate approaches to identify the computational methodology, adaptability and performance across diverse downstream tasks. To address this, these Instruction Fine Tuning and Prompt Tuning paradigms have merged to optimize the model behavior in a complementary way.

Instruction Fine Tuning

The work by Longpre et al. (2023) [1] explores the design decision of publicly available instruction tuning methods using Flan 2022 collection. This study approached the Instruction Fine Tuning framework with mixture of datasets including FLAN, Super-Natural Instructions, P3 for task balancing across diverse natural language understanding and generation tasks. Large Language models including PaLM, Flan-T5 and ChatGPT are used for the model training process to unlock new capabilities in performing to follow human instruction and unified dataset boosts performance across unseen tasks from an instruction. In this work focusses the results of combining Flan 2022 with PaLM 540B to identify the individual factors and comparing size and consistency that leads to comprehensive and reproducible outcome over prior FLAN. Major point of this work is testing the model adding mixed prompts setting to identify the model's performance. However, these methods result in a more competitive starting point for generalizing new instructions on a single new task and still relieve in full model updates and do not incorporate parameter-efficient methods such as prompt tuning. This highlights the needs of hybrid framework that combining instruction fine tuning benefits with the efficiency of prompt-based approaches.

The work about [2] identified the relationship between prompt tuning and instruction fine tuning by Mathew and Ko including the study of "Instruction Fine Tuning: Does Prompt Loss Matter?". The authors examined the study of analyzing the effects of various prompt loss token weights (PLW) for supervised instruction fine-tuning (SIFT) and suggested that using a small non-zero PLW can help stabilize learning when fine-tuning on short-completion data. This approached better results on multiple-choice and short generation benchmarks. By analyzing how loss different loss formulations impact instruction following behavior of the study revealed that optimizing 'prompt components with human instructions and enhances zero-shot generalization on unseen tasks. This highlights the synergistic potential between instruction tuning and prompt optimization, suggesting that effective model alignment may depend on balancing both components. However, this study recognized the overlap concept, it did not formalize a unified hybrid fine tuning framework. This gap underscores the research area for designing a comprehensive hybrid approach that integrates adaptability of prompt level and instruction level generalization withing a single training pipeline.

The work of [3] by Yu et al. (2024) proposed a Progressive LoRA framework (ProgLoRA), which contains a progressive LoRA pool and trains a new LoRA block for each incremental task to reduce knowledge interference specifying two key mechanisms including task-aware allocation for effectively leveraging acquired knowledge at current task and task re call for realigning the model with learned tasks. The experiment optimize that ProgLoRA outperforms MoELoRA and LoRA across two MLLM

backbones on CoIN benchmark, offering a more efficient and stable MCIT approach. This method addresses the challenges of continual instruction tuning in multimodal large language models (MMLMs) which often suffer from catastrophic forgetting and task interferences when models are sequentially fine-tuned on multiple instruction datasets. This emphasizes the need for ProgLoRA implementing task-aware allocation and task recall mechanisms to promote knowledge transfer and mitigate forgetting by achieving improved stability and efficiency. However, this method effectively enhances instruction-based continual learning, it remains limited to LoRA-based adaptation and does not explore prompt-level optimization or hybrid integration of prompt and instruction tuning. This highlights a potential avenue for extending such continual learning frameworks toward a hybrid fine-tuning paradigm to achieve robust multi-task and multimodal generalization.

The previous study [11], develops a system for easily mapping natural language tasks into human readable prompted forms to test the possibility of zero-shot generalization instead be directly induced by explicit multitask learning. This study converts a large set of supervised datasets, which allows benchmarking the ability of a model to perform completely held-out tasks. On multitask collection covering a wide variety of tasks the fine tuning process is applied by a pretrained encoder-decoder model. The model attains strong zero-shot performance on several standard datasets, often outperforming models up to $16\times$ its size.

The work on [12] is about exploring a simple method for improving the zero-shot learning abilities of language models. This study shows that instruction tuning language models on a collection of datasets through instructions significantly improves zero-shot performance on unseen tasks. This work takes a 137B parameter pretrained language model and instruction tune it on over 60 NLP datasets verbalized via natural language instruction templates. Then this instruction tuned model is evaluated also known as FLAN, on unseen task types. FLAN substantially improves the performance of its unmodified counterpart and surpasses zero-shot 175B GPT-3 on 20 of 25 datasets. FLAN, compares favorably against GPT-3 and signals the potential ability for language models at scale to follow instructions.

Prompt Tuning

The work by X. Liu *et al* (2022) [4] presented a novel empirical finding that properly optimized prompt tuning can be universally effective across a wide range of model scales and NLU tasks. It matches the performance of finetuning while having only 0.1%-3% tuned parameters. This study used the GLUE, SuperGLUE and LAMBADA datasets using the GPT T5 model for the experiments to optimize the P-tuning v2 a prompt tuning method, while keeping the main model frozen. The authors identified that it contributes to a novel finding that prompt tuning can be comparable to fine-tuning universally across scales (from 330M to 10B parameters) and tasks with high accuracy and parameter efficiency. However, the study mainly evaluated task-specific scenarios without considering instruction generalization or multi-task adaptation. This limited the need for future exploration into hybrid frameworks to achieve both efficiency and generalization across unseen tasks.

In the study of Li *et al.* (2025) [5] introduced Adaptive Decomposed Prompt Tuning (ADePT), which is composed of a short soft prompt and a shallow token-shared feed-forward neural network that utilizes the token-shared feed-forward neural network to learn the embedding off sets for each token, enabling adaptive embedding offsets that vary according to the model input and better optimization of token embedding offsets. This study enables ADePT to achieve superior adaptation performance without requiring more inference time or additional trainable parameters compared to vanilla PT and its variants.

This method enhances the expressiveness and adaptability of large language models and as a result, this combined short soft prompt embeddings that leads to introduce of hybrid architecture combining a short soft prompt with a lightweight feed-forward neural network. To optimize this study, the PEFT methods including LoRA and standard prompt tuning are used, across various benchmarks. This ADePT method provides faster inference speed with fewer parameters while maintaining competitive accuracy. However, this adaptation reached a strong task-specific efficiency, it focuses on structural prompt decomposition than the higher-level instruction comprehension. This limitation can be an opportunity to enhance the adaptive prompt mechanism using hybrid fine tuning framework with semantic adaptability in large scale instruction following models.

Previous study [9] discusses that prompt tuning is competitive technique for adapting frozen pretrained language models to downstream tasks. The soft prompts are learned through back-propagation and has the ability to incorporate signals from any number of labeled examples. This study shows that prompt tuning grows increasing competitive according to scale as models exceed excessive amounts of parameters, this method “closes the gap” and matches the strong performance of model tuning. This study can be displayed as a simplification of the previous study ‘Prefix Tuning’ [10].

The study [10] propose prefix tuning which is light weight substitute for natural language task generalization. In this study optimizes a small continuous task specific vector called “prefix” while the language model parameters remain frozen. This study applies prefix tuning for GPT-2 for table-to-text generation and to BART for summarization. This study finds that by learning only 0.1% of the parameters, prefix-tuning obtains comparable performance in the full data setting, outperforms fine-tuning in low-data settings, and extrapolates better to examples with topics unseen during training.

Hybrid / Combined Prompt Tuning

Zhu et al. (2024) [6] introduced the novelty prompt tuning method the Instruction Aware Prompt Tuning (IAPT) method for the parameter-efficient fine tuning for LLMs. The study utilized super-natural instruction and Alpaca datasets with GPT and T5 models to evaluate its effectiveness. The authors addressed a key limitation of Soft Prompt Tuning which is many tokens must be inserted into the input sequences which considered as soft prompt tuning is less than Low-rank adaptation (LoRA) in the LLM era. The proposed IAPT method significantly reduced this burden by requiring four soft tokens improving the performance of prompt tuning on instruction datasets. Moreover, the study identifies that the IAPT method is more efficient than LoRA under the single-backbone multi-tenant settings. Also, this proposed that 3 downstream performances include self-attention pooling, learning different activation and cross layer parameters. However, while IAPT successfully integrates instruction context into prompt generation, it remains limited to prompt-based adaptation and does not employ instruction fine-tuning as a full training paradigm. This leaves open the potential for a hybrid framework that combines the efficiency of IAPT like prompt mechanisms with the generalization capabilities of instruction fine tuning.

The work on [7] by Li et al. (2025) introduced the Free-From Instruction oriented Prompt Optimization (FIPO), which allows precise optimization of the core task instructions in naive prompts in a model-agnostic manner. Unlike conventional prompt tuning methods that reply on fixed or manually engineered templates, FIPO method employs instruction-oriented templates combined with preference-based learning, allowing the model to dynamically optimize prompts based on human instruction, feedback and task context. This work used the models of LLAMA-2 and Flan-T5 and collected a large-scale prompt optimization preference dataset, employed multiple fine-tuning strategies, and then

validated the efficacy across objective benchmarks with various downstream generators, by finding the combination of prompt optimization and modular finetuning yields better performance alignment to enable more flexibility between model instructions and downstream task requirements, improving both instruction-following behavior and response coherence. Furthermore, FIPO’s modular design supports fine-grained control over prompt components, bridging aspects of instruction tuning and prompt optimization. However, despite its hybrid characteristics, FIPO still treats instruction and prompt optimization as loosely coupled modules rather than an integrated training framework. This suggests an opportunity to explore the gap of unified hybrid fine tuning model by joining instruction level understanding of prompt -level efficiency.

In the study [8] by M. et al. (2023) introduced hybrid prompt framework with bridging language models and human priors in prompt tuning for VQA to mitigate the gap between upstream pre-training tasks and downstream VQA tasks. Experimented on commonly used VQA v2 dataset demonstrates the effectiveness of Hybrid Prompt, showing that it outperforms previous VQA methods and obtains new state-of-the-art results. This work aims to answer the natural language question about a given image by understanding multimodal content. The study addressed the limitations of traditional prompt tuning methods, which rely on fixed-length soft prompts or hard text information. This hybrid design allows the model to leverage human priors for contextual grounding while maintaining flexible and optimization efficiency through learnable parameters. Experimental evaluations demonstrate that HybridPrompt achieved superior performance over standard prompt tuning baselines on multiple VQA benchmarks, highlighting the benefits of integrating human priors into prompt design. However, the framework’s optimization remains primarily prompt driven, lacking instruction-level generalization across diverse language tasks. This indicates the potential needs of further extend of hybrid design toward instruction aware hybrid fine tuning frameworks that unify both prompt and instruction learning broader task adaptability.

the previous study [13], presents a method for aligning LLMs with user intent on across a variety of tasks by fine-tuning with human feedback. this process starts with a set of labeler-written prompts and prompts submitted through the OpenAI API, which then collected into a dataset of labeler demonstrations of the desired model behavior, which used to fine-tune GPT-3 using supervised learning. Then a dataset of rankings of model outputs is collected, which then used to further fine-tune this supervised model using reinforcement learning from human feedback. the resulting model is called “InstructGPT”. InstructGPT models show improvements in truthfulness and reductions in toxic output generation while having minimal performance regressions on public NLP datasets.

2.2 Challenges and Limitations

Although the work in [1] optimizes a significant contribution by introducing the FLAN 2022 collection, however, there are several limitations remaining, focusing on instruction based fine tuning without exploring integration with prompt tuning or parameter-efficient techniques, which shows strong potential for low-resources model adaptation. Also, this work needs to be relied on large scale models like PaLM 540B and Flan-T5 XXL which make the framework more computationally expensive and less feasible for smaller applications.

While the study of [2] provided valuable insights into optimizing loss ratios in instruction-based model adaptation, there are several limitations remaining. Since the authors used pre-trained models and no layers were newly initialized, there was little variance in initial experiments which limited runs to a

single seed of 42. This leads not to capturing behavior in diverse training conditions or model architecture. Due to the selection of PLW thresholds and the definition of short versus long completions leads to lacking strong theoretical justification or broader empirical validation. Also, this relied on LLM as evaluator method which, bias compared to standard human or default auto-evaluators. Overall, the study focuses solely on loss balancing within instruction fine-tuning and does not explore how such weighting could interact with prompt tuning or hybrid fine-tuning frameworks.

[3] To validate the effectiveness of MCIT methods, future work could focus on creating more comprehensive and challenging benchmarks that encompass a wider array of tasks and domains. This method is influenced by the task order to some extent that maintaining robustness across different task orders is a valuable research direction. For instance, further enhancing the ability to transfer knowledge could allow the model to not only learn new knowledge but also revisit and fill gaps in previously learned knowledge. Also, consider that the total parameters of the model will increase as new tasks are added. Since the additional parameters introduced by our method at each task come from one LoRA block, which is small compared to the total number of model parameters, the impact of the increasing number of tasks. The authors identified that some methods, like the fusion of similar block fusion and block pruning, can be used to address this problem. Additionally, this work does not highlight any research gap of unifying adaptive prompt tuning with instruction based fine tuning to achieve scalable generalization.

The work on **[4]** which is Prompt tuning v2 has notable these limitations that the evaluation is conducted on task specific benchmarks like large language datasets including GLUE and SuperGLUE rather than instruction style tasks limiting its applicability to follow human instructions better. This work mainly focuses on prompt tuning without integrating instruction fine tuning or adaptor methods, which does not adapt well to unseen instructions. Also, this study depends on prompt embedding design, which is not smoothly scaled to highly diverse domains without manual tuning. Although this work did not address how prompt embedding might be combined with efficient instruction level fine tuning with hybrid framework.

Limitations of the work **[5]** highlighted that evaluation focus only on token-level downstream NLP tasks, low understanding of how the method generalizes to follow or generate tasks where prompt compositions are critical. Although the adaptive feed forward layer reduces the inference speed advantage offered by lightweight prompt tuning technique. Also, this study does not explore combining with instruction tuning or hybrid adaptation methods, where possible missing potential synergies between decomposed prompt and instruction level supervision. Theoretical analysis emphasizes expressiveness but lacks extensive empirical validation across diverse model architectures and domain settings.

The work of **[6]** shows limitations where the limited computational resources acknowledging super-sized open sourced LLMs and L1aMA-2 13B and 70B are not experimented due to limited computation resources. Also, information extraction is not considered.

In the work of **[7]** demonstrates several limitations considering overwhelmed cheating notes that provides overly detailed instructions that can be considered as “cheating notes”. This issue is particularly prevalent in tasks involving mathematical calculations and it may not align with the intended use of prompt optimization. The second limitation mentioned is Evaluation Metrics. In here the whole work must be consider whole aspects like fairness and ethical implications. Another

limitation of this work is FIPO does not include optimization of the in-context examples, but rather only focuses on the optimization for the task instructions.

This work [8] identifies the limitations of processing instruction- following tasks remain untested for multimodal tasks including images and texts. Authors identified that the prompt design depend on human priors like hard prompt templates, therefore thus method’s performance may degrade when human priors are not aligned with new domains or when not available. This framework optimized the lack evaluation in cross-task or cross domain settings, limiting its generalized demonstrations. The hybrid prompt does not incorporate instruction fine tuning which remains fundamentally prompt driven approach rather than hybrid tuning.

2.3 Comparison Table – Literature Summary

Paper	Method	Datasets	Models	Finding	Strengths	Limitaions
[1] The Flan Collection - Longpre et al (2023).	Instruction Fine-Tuning Framework with mixture-of-datasets and task balancing	FLAN, Super-Natural Instructions , P3	Flan-T5, Flan-PaLM	Unified dataset boosts performance across unseen tasks	Comprehensive and reproducible improved over prior FLAN	No analysis of combining prompt tuning and instruction tuning
[2] Instruction Fine-Tuning: Does Prompt Loss Matter? (2024)	Examines prompt-related loss functions within instruction fine-tuning frameworks	FLAN, Alpaca, Dolly	T5, LLaMA	Prompt loss design affects model alignment and stability during instruction tuning	Reveals importance of prompt signals in instruction fine-tuning	Does not propose a new method; lacks hybrid optimization pipeline
[3] Progressive LoRA for Multimodal Continual Instruction Tuning (2025)	Progressive LoRA — incremental LoRA block allocation for continual multimodal instruction tuning	CoIN, MM-Instruct	Multimodal LLMs (BLIP-2, Flamingo)	Outperformed standard LoRA and MoE-LoRA in continual learning stability	Effective for multimodal continual instruction tuning	Limited to LoRA-based parameter tuning; lacks prompt-instruction hybridization

[4] P-Tuning v2: Prompt Tuning Can Be Comparable to Fine-tuning Universally Across Scales and Tasks (2022)	P-Tuning v2 — reparameterized prompt tuning for scalability and universality	GLUE, SuperGLUE, FewGLUE	GPT-2, BERT, RoBERTa	Matched full fine-tuning performance with fewer parameters	Parameter-efficient and task-general; strong empirical validation	Does not address instruction integration; limited exploration of hybrid synergy
[5] ADePT: Adaptive Decomposed Prompt Tuning (2025)	Adaptive decomposition of prompts into modular components for efficient tuning	Super-Natural Instructions, OpenAI Benchmarks	T5, GPT-J	Improved parameter efficiency and adaptability using decomposed prompt layers	High flexibility, task-wise adaptability	Focused only on prompt optimization; no instruction or hybrid integration
[6] IAPT: Instruction-Aware Prompt Tuning for Large Language Models (2024)	Instruction-Aware Prompt Tuning (IAPT) — integrates instruction signals into soft prompt tuning	Super-Natural Instructions, Alpaca	GPT, T5	Achieved efficiency using only 4 soft tokens; improved prompt tuning over LoRA in multi-tenant settings	Combines instruction awareness with prompt tuning; efficient token use	Limited to soft prompt representation; lacks deeper hybridization or adaptive mechanisms
[7] FIPO: Free-form Instruction-oriented Prompt Optimization (2024)	Modular hybrid tuning combining instruction templates + preference-based prompt optimization	Preference Dataset (custom)	T5, GPT variants	Higher adaptability and modularity across free-form instruction settings	Introduces modular hybrid idea connecting prompt and instruction learning	Not a full hybrid fine-tuning pipeline; limited exploration of multi-task generalization

[8] Hybrid Prompt: Bridging Language Models and Human Priors in Prompt Tuning for Visual QA (2023)	HybridPrompt Framework — combines human priors and learned prompts for VQA	VQA v2, GQA	VisualBERT, ViLT	Improved visual reasoning accuracy and robustness by fusing human and learned priors	Strong hybrid strategy showing prompt + external prior synergy	Focused on visual QA; lacks generalization to NLP tasks
[9] The Power of Scale for Parameter-Efficient Prompt Tuning (2021)	Instruction Aware Prompt Tuning (IAPT) – integrates instruction signals into soft prompt tuning	GLUE, SuperGLUE Natural questions	T5, GPT-2	Found out that continuous prompts is more suited for large models with task generalization compared to full fine tuning	Avoids catastrophic forgetting, highly parameter efficient,	Suitable for larger models Limited interpretability Limited task generalization
[10] Prefix-Tuning: Optimizing Continuous Prompts for Generation (2021)	Adaptive decomposition of prompts into modular components for efficient tuning	E2E, WebNLG CNN/Daily Mail	GPT-2, BART	Effective tuning with minimal parameters due to Prefix prompts	Efficient, lightweight, flexible task generalization	Lacks the instruction based learning , Focused only on generative NLPs
[11] Multitask Prompted Training Enables Zero-Shot Task Generalization (T0) (2025)	Instruction Fine Tuning framework with the combination of datasets and task balancing	P3, SuperNatural Instruction	T5 , T0	Strong zero-shot generalization realized through Multitask prompted instruction tuning	Combines multitask and instruction tuning and improves the cross domain performance	Computationally intensive Limited parameter reusability

[12] Finetuned Language Models Are Zero-Shot Learners (2021)	Examine the prompt related loss functions within instruction fine tuning frameworks	P3, Natural instructions	PaLM, T5	Improves the reasoning ability and zero-shot task transfer through instruction fine tuning	Increased task generalization Compliance to human instructions	Less parameter efficiency, High model traing cost
[13] Training Language Models to Follow Instructions with Human Feedback /InstructGPT (2022)	Hybrid tuning combining instruction templates + preference based prompt optimization	Human feedback, InstructGPT datasets	GPT-3	Combines human preference optimizatio n with instruction templates for safer, aligned output.	Improves alignment, interpretabili ty, and instruction- following behaviour.	Expensive to train Challenging to adapt into new domains.

3. Identified Research Gap

Among the rapidly evolving landscapes of large language model (LLM) adaptation, methods such as Instruction Fine-Tuning (IFT) and Prompt Tuning (PT) have emerged as powerful methods to turbocharge model performance. Though both offer mammoth improvements separately, the enormous chasm lies in consolidating these approaches to create a detailed and optimized framework. The major gap is the lack of rigorous exploration of how Instruction Fine-Tuning and Prompt Tuning can be combined in the optimal way to improve performance, generalization, and scalability. An in-depth exploration of the relationship between these approaches, especially in the scenario of multi-task, multi-modal, and zero-shot learning, is not adequately explored.

Below is a detailed breakdown of the research gaps based on the reference papers:

1. Absence of Unified Hybrid Fine-Tuning Framework

A significant gap that can be identified across several papers, for example, [2] and [6], is that of not having a single framework that integrates Instruction Fine-Tuning along with Prompt Tuning within a single method. While IAPT (Instruction-Aware Prompt Tuning) introduces instruction signals as part of soft prompt tuning, such methods are still only good for prompt tuning alone or instruction fine-tuning alone without fully addressing how both can be combined within a hybrid framework. As a point of example, IAPT (Zhu et al., 2024) introduces instruction signals as part of prompt tuning without trying to venture further into hybridization where instruction fine-tuning and prompt tuning are both optimized over several levels of models. Studies therefore are mainly based on optimizing single components, as opposed to delving into a complete pipeline that can generalize over several tasks as well as several domains within a single method seamlessly.

This gap is especially significant since both methods—prompt tuning and instruction fine-tuning—have unique strengths. Prompt Tuning is extremely parameter-efficient, minimizing trainable parameters, whereas Instruction Fine-Tuning makes task generalization possible through instruction following in natural languages that models are trained on. A combination of both methods may provide models that take advantage of parameter usage while possessing instruction fine-tuning's task-generalization ability. Yet no research has methodically explored how it is best to sequence or combine these two approaches for maximum impact under regularized resources.

2. Task-Specific Generalization and Multi-Task Performance

The lack of thorough investigation into how hybrid approaches can achieve better multi-task generalisation than either approach alone is a second significant gap, as covered in papers such as [4] and [6]. P-Tuning v2 (Liu et al., 2022) and IAPT (Zhu et al., 2024) are two recent studies that show good performance in single-task or single-domain tasks, but they are not very good at assessing how a hybrid framework can generalise across different tasks, domains, and languages.

For instance, by using fewer parameters to accomplish prompt tuning that matches full fine-tuning performance, P-Tuning v2 achieves impressive efficiency. It doesn't, however, investigate how instruction signals might improve multi-domain or task-specific generalisation. Similar to this, IAPT achieves high efficiency in multi-tenant settings, but it places less emphasis on how instruction-

following skills could be combined to improve cross-task performance and more emphasis on prompt optimisation.

One major drawback for models designed to be used in dynamic, real-world situations is their incapacity to generalise across a variety of tasks. It may be possible to handle a wider variety of tasks without requiring a lot of retraining if models are able to both follow instructions (instruction fine-tuning) and modify their input prompts (prompt tuning) for particular tasks.

3. Scalability of Hybrid Methods for Large-Scale Models.

Another important gap in the literature is scalability. The scalability of combining prompt tuning and instruction fine-tuning at large scales is not sufficiently addressed in papers like [5] and [6], which show the parameter efficiency of prompt tuning, particularly in models like GPT-J and T5. When it comes to large-scale models with billions of parameters, such as Flan-T5 or PaLM, techniques like ADePT (Tang et al., 2025) and P-Tuning v2 do not scale well, despite their potential in parameter efficiency.

To manage the massive amounts of memory and processing power required for fine-tuning, large-scale language (LLMs) models require effective training techniques. A more scalable solution might be provided by the hybrid fine-tuning strategy, which combines prompt and instruction tuning; however, it is still unclear how to scale these techniques without sacrificing effectiveness and performance. In particular, how to maintain performance across tasks while optimising both prompt-tuned and instruction-tuned parameters to fit within the constraints of contemporary computational resources.

4. Instructional Complexity and Cross-Domain Applications.

The difficulty of modifying hybrid models for cross-domain and cross-lingual tasks represents the last gap, as covered in papers such as [8] and [7]. It has been demonstrated that adding human priors to frameworks such as HybridPrompt can improve model performance for Visual Question Answering (VQA). However, these models do not generalise well to NLP or other multimodal applications, and they are frequently restricted to visual QA tasks.

Cross-lingual adaptation and domain-specific complexities (such as task-specific language and technical jargon) must be taken into account when combining prompt tuning and instruction fine-tuning. Without sacrificing its ability to generalise, a hybrid system must handle linguistic diversity and domain-specific subtleties. How well can a domain-adapted hybrid model function, for instance, in different languages or on tasks that are outside of its domain? This question remains under this research.

Summary of Research Gaps:

- 1. Unified Hybrid Framework:** To optimise task generalisation and parameter efficiency across various tasks and domains, a systematic hybrid framework that integrates prompt tuning and instruction fine-tuning is lacking.
- 2. Multi Task and Multi-Domain Performance:** Since single-task models are frequently the focus of current research, there is a lack of models that can manage multi-task learning and domain-specific adaptations through the use of hybrid fine-tuning.

3. **Scalability of Hybrid Methods:** Although both approaches are efficient in terms of parameters, little research has been done on how they can scale well for large models and continue to be effective in settings with limited computational resources.
4. **Cross-Domain and Cross-Lingual Generalisation:** Future research must focus on models that can generalise to new domains and languages without requiring a great deal of retraining, as current hybrid approaches are usually restricted to particular domains or tasks.

4. Future Directions

The following future directions can help address the issues and unlock the potential of combining Instruction Fine-Tuning (IFT) and Prompt Tuning (PT) in a hybrid manner for large language models (LLMs) based on the identified research gaps:

1. Development of Unified Hybrid Fine-Tuning Framework

Future research must focus on creating a systematic pipeline that integrates Prompt Tuning (PT) and Instruction Fine-Tuning (IFT) at various model levels. Because existing approaches, such as IAPT, usually focus on either of these strategies separately, there is a significant knowledge gap about how prompt tuning and instruction fine-tuning can be combined to achieve better outcomes. Future studies could look at the order of operations that produces the best model performance, as well as the sequential or parallel application of prompt tuning and instruction fine-tuning. To test whether instruction-following improves prompt adaptability, for example, prompt tuning could be used to provide task-specific signals before instruction fine-tuning. Future models could reduce the number of parameters needed while improving generalisation across various tasks and domains by filling this gap.

2. Multi-Task and Multi-Domain Generalization

The study of how hybrid fine-tuning techniques can enhance multi-task and multi-domain performance is another crucial avenue for future research. Existing models, like P-Tuning v2 and IAPT, perform well in single tasks or domains but frequently fall short in generalising to different tasks or multi-domain environments. A hybrid system might enable models to handle a variety of tasks without the need for retraining or domain-specific modifications by fusing the flexibility of prompt tuning with the instruction-following capabilities of instruction fine-tuning. The development of models that function well in zero-shot and few-shot learning environments should be the main goal of future research. This would create opportunities for models that automatically learn new languages or tasks, which is crucial in real-world situations where task diversity is unavoidable.

3. Scalable Hybrid Fine-Tuning for Large-Scale Models

Although both prompt tuning and instruction fine-tuning have demonstrated parameter efficiency for smaller models, scaling such methods for a large model—such as Flan-T5 or PaLM—represents a challenge. Future work should examine how to efficiently scale hybrid fine-tuning approaches to operate on an LLM of billions of parameters without compromising performance nor computational efficiency. For example, mixed-precision training, model pruning, and distributed fine-tuning could be important for developing hybrid fine-tuning approaches for large-scale models. This research will need to take the trade-offs of computational resources and model performance into consideration, while ensuring both prompt and instruction signals are being appropriately leveraged at scale. Additionally, it will be important to determine how to scale the hybrid approach while maintaining performance across a number of tasks to enhance the real-world usability of the model.

4. Cross-Domain and Cross-Lingual Generalization

In addition to general task performance, a hybrid fine-tuning methodology should be adaptable in cross-domain and cross-language contexts. This can be especially relevant when considering expert domains, such as legal, medical, or scientific domains as they require domain-specific jargon, phrasing and nuance. A hybrid approach that combines prompt tuning with instruction fine-tuning may allow for

such adaptation by having a model consider whether it should adjust its prompts based on the domain and whether it is appropriately following the instruction. Further, cross-lingual adaptation is another looming concern. Future work should examine how hybrid fine-tuning systems can transfer instruction following ability to languages other than English while maintaining performance as much as possible, and explore how these fine-tuning approaches can be utilized in low-resource languages. Understanding how these hybrid processes can generalize across both domain and cross-linguistic contexts is critical to developing our models to be robust and adaptable.

5. Hybrid Fine-Tuning for Multimodal Tasks

As multimodal models become more common, it will be important to understand how hybrid fine-tuning can be utilized on tasks that support both text and non-text modalities, such as visual and auditory modes, etc. Although multimodal systems, such as VQA models, have shown we can successfully integrate human priors with ML prompts, we have only begun a process of understanding how they will come together within hybrid fine-tuning tasks on multimodal data with instruction-following and prompting prompts capable of either providing a unified structure that adds prompt capacities for text domain-specified prompt tuning or a distinction where instruction-following and prompt models can either be formed sequentially or concurrently. Future advancements in our understanding of multimodality, hybrid tune, and prompt optimization models, could explore possibilities to optimize models on either instruction-following and prompt optimization or both focused either entirely or with supportive prior associated prompt tuning. Whether hybrid fine-tuning involves images, sounds, or visual inputs, the challenge will be to include prior context, integrate multiple modes, and optimize the hybrid fine-tune process while maintaining both instruction-following and prompt generalization capabilities.

6. Customized and Domain-Adaptive Hybrid Systems

Personalization and domain adaptation is another area of research that has great potential for hybrid fine-tuning. In many real-world settings, models would ideally be customized to the user or domain. For example, a virtual assistant or customer service bot has to adjust to user preferences and take into consideration domain-specific lexicons. A hybrid fine-tuning system could be developed to consider both personalization and domain adaptation by adapting to the user's prompt representations, while maintaining the general ability of the model to generate task instructions. In future research, we could explore adapting instruction fine-tuning to a specific user's context, while prompt tuning allowed the system to adapt to specific tasks (e.g., giving medical advice, financial recommendations). Examining this and other hybrid approaches will aid in creating models that may be adapted across domains and continually develop based upon user interaction.

7. Evaluation of Hybrid Models in Real-World Applications

Finally, to make hybrid fine-tuning techniques of practical value, it is important to test them on real-world tasks. This involves testing models in dynamic situations, such as customer support, healthcare, or content creation, in which models need to keep adapting to new directions, user inputs, and changing tasks. Testing hybrid models in such situations will give us an idea about their performance in real-world scenarios, especially in terms of robustness, flexibility, and efficiency. Long-term observation of hybrid fine-tuned models' performance in the long term is what needs to be studied in order to know whether or not they can meet real-world needs without retraining every time. This will be crucial in determining whether these models are viable for large-scale use and high-demand applications.

5. Conclusion

In summary, although there has been substantial progress in Instruction Fine-Tuning (IFT) and Prompt Tuning (PT), the coordination of both into a singular and effective framework presents a noteworthy and lasting gap in research. The unexploited option that exists is hybrid fine-tuning—a method that combines the benefits of the two approaches. There is no one-neural pipeline for instruction-following and prompt optimization, therefore limiting the scaling and generalization of models, including across tasks, domains, and languages.

The research gaps that have been identified show that future research needs to create scalable hybrid systems which maintain operational efficiency and achieve high performance results for handling extensive real-world tasks with reduced computational requirements. The development of hybrid methods for multi-task learning and cross-domain adaptation and multimodal tasks will determine the future capabilities of language models. The combination of personalized domain-adaptive and cross-lingual features in hybrid systems creates potential for building more robust models which need minimal retraining to adjust to new challenges.

The combination of prompt tuning with instruction fine-tuning will establish new standards for language models by creating AI systems that work more efficiently while understanding different situations and adapting to new tasks. The future direction of hybrid fine-tuning requires researchers to use these synergies for developing models that perform well across multiple languages and domains and tasks which will enable them to operate effectively in practical real-world scenarios.

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